Titouan Brunel

Student at MVA, ENS Paris-Saclay & CentraleSupélec

99 rue du Bac, 75007
Paris, France
(+33) 6 15 01 50 55
titouan.brunel@student-cs.fr
titouan-brunel
Citizenship : French

Education

2026 – 2027 **Master 2 MVA (Mathematics, Vision, Learning) at École Normale Supérieure Paris-Saclay**
Courses: Optimal Transport · Diffusion and Flow Matching · Computer Vision · Object Recognition · LLMs · Deep Learning · Graphical Models · Optimization · Reinforcement Learning

2023 – 2027 **Student at CentraleSupélec, Paris-Saclay University**, Ranked 2nd in Academic Ranking of World Universities in Mathematics, Top 3% of 1000 students.

Mathematics:
- Convergence, Measure Theory & Probability
- Functional Analysis
- Statistics
- Stochastic Processes

Computer Science:
- Mathematics for Data Science
- Machine Learning
- Statistical Learning Theory
- AI and Deep Learning

2021 – 2023 **Preparatory Classes (PCSI/PC*)**, *Lycée Montaigne, Bordeaux, France*
- **Eligible to all top French engineering and science schools** (École Normale Supérieure Ulm, École Polytechnique, Centrale-Mines network)
- Two-year intensive program in Mathematics, Physics and Chemistry for Grandes Écoles entrance exams

Professional Experience

Mar 2026 – Aug 2026 **Research Intern (6 months)**, *at Mila, Québec AI Institute*, Montréal, Canada
Research on mapping deep learning molecular representations (AF3, Boltz-2, 3D GNNs) to LLM-encoded neuroscience effect representations, and conditional flow matching for novel psychoactive drug generation conditioned on encoder internal representations. *Supervised by Prof. Danilo Bzdok, MD, PhD*

Aug 2025 – Feb 2026 **Quantitative Researcher Intern (6 months)**, *at Edo Theory*, Paris, France
Led research on percolation theory to model cryptocurrency limit order book dynamics, and developed deep learning architectures for non-linear time series prediction and trading signal generation. *Supervised by Grégoire Macqueron, Founder & Research Director*

Mar 2025 – Sept 2025 **Machine Learning Engineer**, *FoodTech Startup*, Paris, France
Developed and deployed an end-to-end sales forecasting system handling sparse data through hierarchical models (individual, cluster, and global levels). Implemented intelligent clustering and probabilistic forecasting with external data integration for anomaly detection. System currently running in production.

Sept 2024 – June 2025 **BSc Physics Teaching Assistant and Oral Examiner (Khôlles)**, *CentraleSupélec*, Paris

Academic & Research Projects

Oct 2026 – Apr 2027 **JEPA World Models for Compiler Optimization**, *CNRS*, Developing a self-supervised predictive world model (JEPA) over LLVM IR to plan optimization pass sequences in latent space, aiming for sample-efficient search with far fewer real compilations than CompilerDream and standard LLVM pipelines.

Sept 2026 **Bayesian Optimization Research Project**, *CentraleSupélec*, Designed novel acquisition strategies to push the Pareto frontier beyond baselines in multi-objective Bayesian optimization. Under Prof. Vazquez.

Apr 2025 **Martingale Optimal Transport for Exotic Options Pricing**, Solved MOT via LP approximation with marginal discretization and entropic regularization (Bregman projections).

Jan 2025 **Machine Learning Competition (2nd)**, *Kaggle*, Classified geographical areas using ML on multi-modal data (position, pixel values, temporal evolution, descriptions), ranked 2nd with 98%+ accuracy.

Interests

Music	Guitar: 7 years, played in 2 orchestra
Sports	Tennis: 15 years
Leadership	Forum CentraleSupélec: 2024 Vice-President
Poker	PokerStars Ambassador of CentraleSupélec
Volunteer	Summer Camp L'Espoir: Board Member

Computer Skills

Programming	Python, SQL
Tools	LaTeX, Git, Linux

Languages

French	Mother Tongue
English	Fluent
Spanish	Intermediate